SpaceX Falcon 9 First Stage Landing Prediction

Capstone project, IBM Data Science Professional Certificate (Applied Data Science Capstone)

Todd Sakai

November 2026

Summary

A Falcon 9 launch is much cheaper than a competitor's launch mainly because SpaceX reuses the first stage. If we can predict whether the first stage will land, we can estimate the cost of a launch. This project collects SpaceX launch data, explores it, and trains four classification models to predict the landing.

- **Data:** 90 Falcon 9 launches from June 2010 to November 2020. 66.7% of the first stages landed.
- **Trend:** the landing success rate rose from 0% in 2010 to 2013 to 90% in 2019.
- **Launch site:** KSC LC-39A has the most successful launches (41.7% of all successes) and the highest success rate (76.9%).
- **Model:** a support vector machine, k-nearest neighbors and logistic regression all predicted 83.3% of the test launches correctly (15 of 18). Each model found all 12 landings, but it also predicted a landing for 3 of the 6 launches that did not land.

Methodology

1. **Data collection.** Launch records came from the SpaceX REST API and from the Wikipedia list of Falcon 9 launches (web scraping with BeautifulSoup). The public SpaceX API was offline in November 2026, so the API notebook uses IBM's saved copy of the API result.
2. **Data wrangling.** Each landing outcome became a label, `Class`: 1 if the first stage landed and 0 if it did not. A missing payload mass was replaced with the mean.
3. **Exploratory analysis** with SQL queries, charts, an interactive Folium map and a Plotly Dash dashboard.
4. **Feature engineering.** Categorical features (orbit, launch site, landing pad and core serial) became one-hot columns, which gave 83 features. The features were then standardized.
5. **Models.** Logistic regression, a support vector machine, a decision tree and k-nearest neighbors were each tuned with a grid search and 10-fold cross-validation on 72 launches, then tested on 18 held-out launches.

Exploratory Data Analysis

SQL

Question	Answer
Launch sites	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A and VAFB SLC-4E
Total payload carried for NASA (CRS)	45,596 kg
Average payload of booster version F9 v1.1	2,928.4 kg
First successful landing on a ground pad	22 December 2015
Most common landing outcome, June 2010 to March 2017	No attempt (10 launches)

Landing success over time

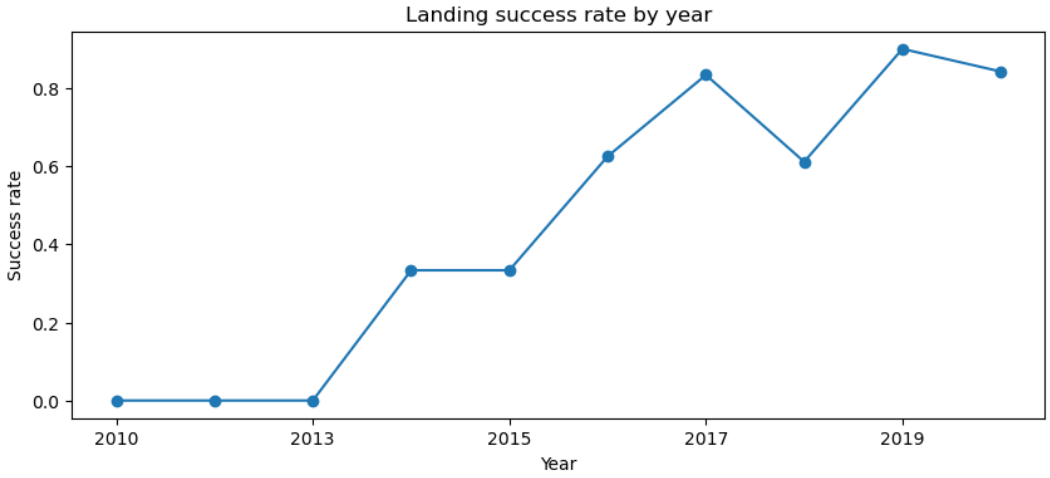


Figure 1: Landing success rate by year

The success rate has risen steadily since 2013, as SpaceX improved the rocket and the landing method.

Flight number and launch site

A scatter plot of flight number against launch site (notebook 5) shows that later flights land more often at every site. Experience matters more than the site itself.

Success rate by orbit

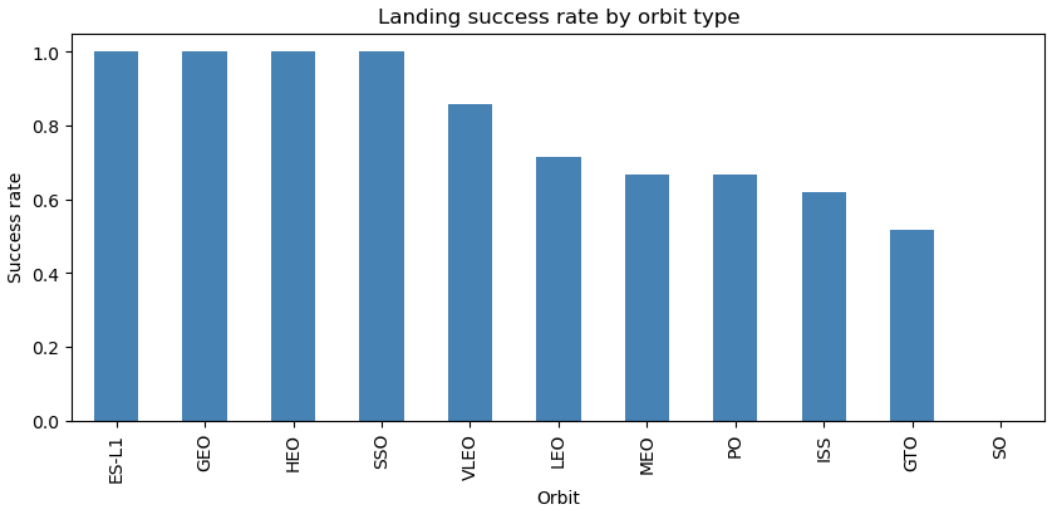


Figure 2: Landing success rate by orbit type

The ES-L1, GEO, HEO and SSO orbits had a 100% success rate, but with only 1 to 5 launches each. GTO, a common orbit for heavy commercial satellites, had a success rate of only 52%.

Interactive Analysis

Launch site map (Folium)

All launch sites are on a coast, so rockets fly over the ocean. CCAFS SLC-40 is 0.51 km from the coastline but 23.19 km from Titusville and 53.43 km from Melbourne, the nearest cities. The interactive maps are in the `maps/` folder of the repository.

Dashboard (Plotly Dash)

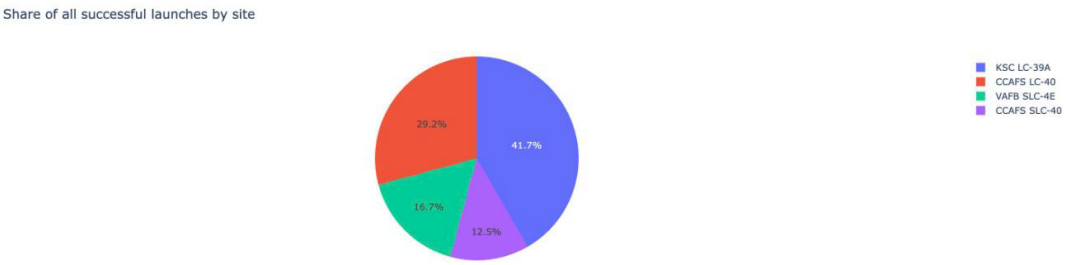


Figure 3: Share of all successful launches by site

KSC LC-39A has the largest share of successful launches, 41.7%.

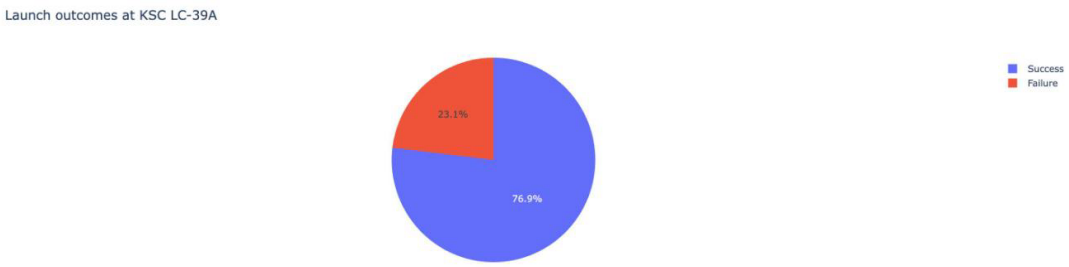


Figure 4: Launch outcomes at KSC LC-39A

KSC LC-39A has the highest success rate, 76.9% (10 of 13 launches).

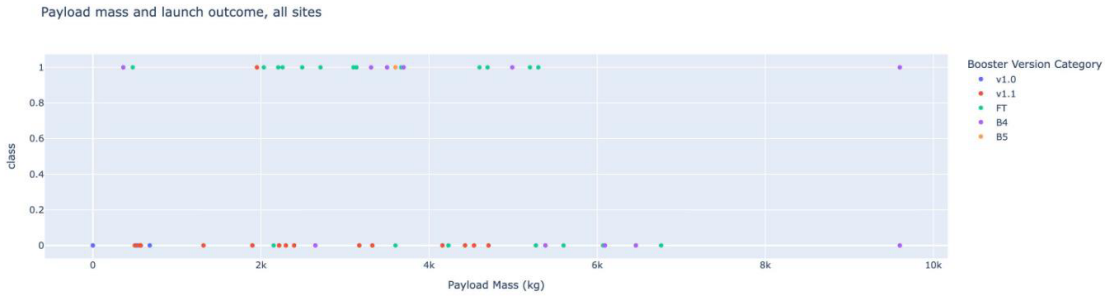


Figure 5: Payload mass against launch outcome for all sites, colored by booster version

The FT booster version has the most successful launches across payload masses.

Predictive Analysis

Model	Validation accuracy (10-fold)	Test accuracy
Support vector machine	0.848	0.833
K-nearest neighbors	0.848	0.833
Logistic regression	0.846	0.833
Decision tree	0.836	0.778

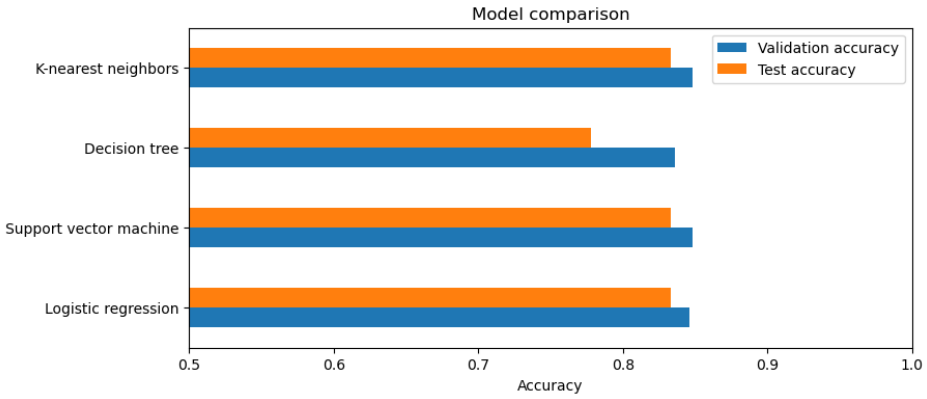


Figure 6: Validation and test accuracy of the four models

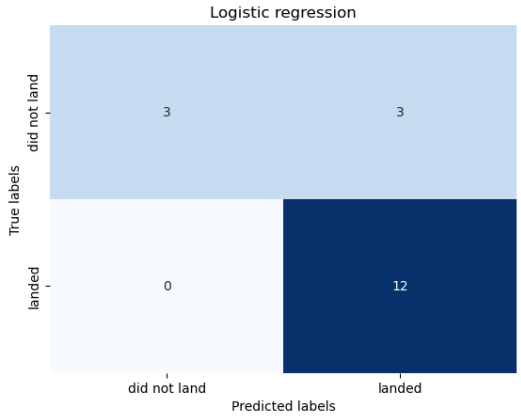


Figure 7: Confusion matrix of the logistic regression model on the 18 test launches

Three models tie on the test set, so the test set alone cannot choose between them: it has only 18 launches. Each of the three found all 12 landings but predicted a landing for 3 of the 6 launches that did not land. These false positives are the main weakness, because a cost estimate based on them would assume that a stage can be reused when it cannot.

Conclusions

- Landing success improved a lot over time, so the flight number is a strong predictor.
- KSC LC-39A is the most successful launch site.
- The ES-L1, GEO, HEO and SSO orbits had the highest success rates, but they had few launches.
- A tuned classifier predicts the landing correctly about 83% of the time. A larger test set, or repeated cross-validation, is needed to choose one model over another.